CUSTOS: AI-Powered Surveillance HUB

Core Feature (1) Asset Protection

Official System Workflow & Documentation

CUSTOS is a next-generation security solution that leverages real-time AI to understand human-to-object interactions. This document provides a structured walkthrough of the system's capabilities, from its vision to its detailed operational interface.

---

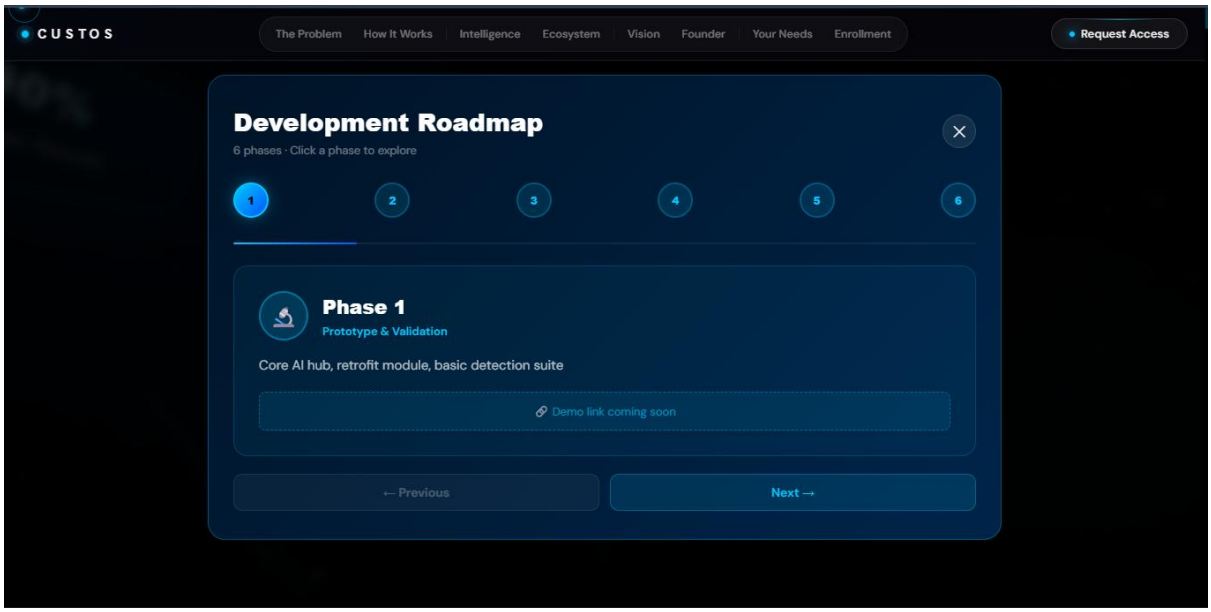
1. Vision and Mission

The CUSTOS platform is designed to bridge the gap between simple video recording and intelligent situational awareness. Its mission is to protect critical assets—from homes to nations—by providing a security system that "thinks" and "understands" rather than just "sees."

---

2. Development Roadmap

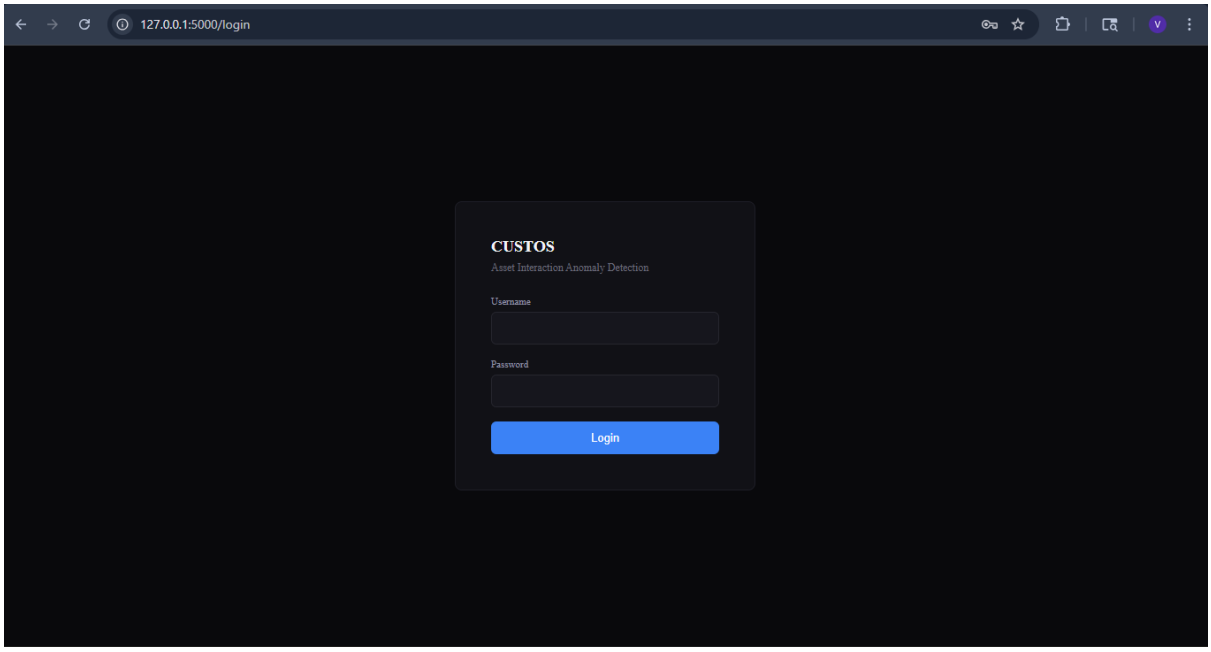
CUSTOS is built on a structured development path. Starting with Phase 1 (Prototype & Validation), the system focuses on establishing a core AI hub, retrofitting existing modules, and deploying a basic detection suite to ensure a solid foundation for advanced security features.



---

3. Secure Access Control

The system is protected by a dedicated authentication layer. Authorized users access the CUSTOS environment through a secure login portal, ensuring that sensitive surveillance data and asset tracking configurations remain confidential.

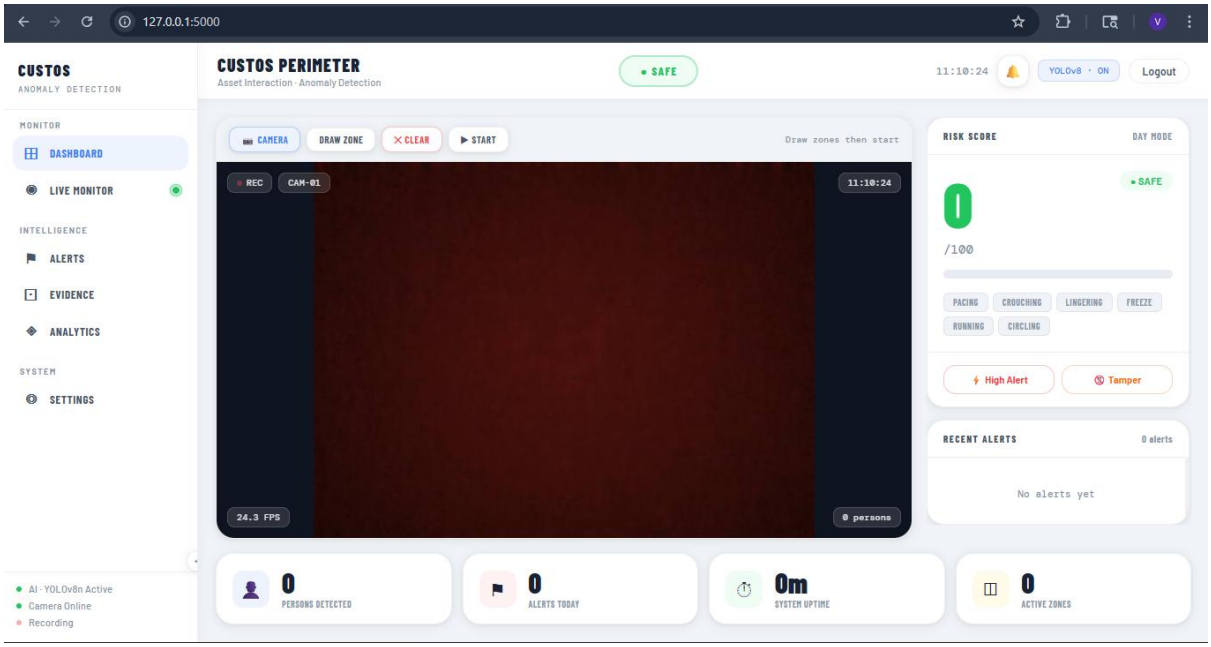


---

4. Central Command: The Dashboard

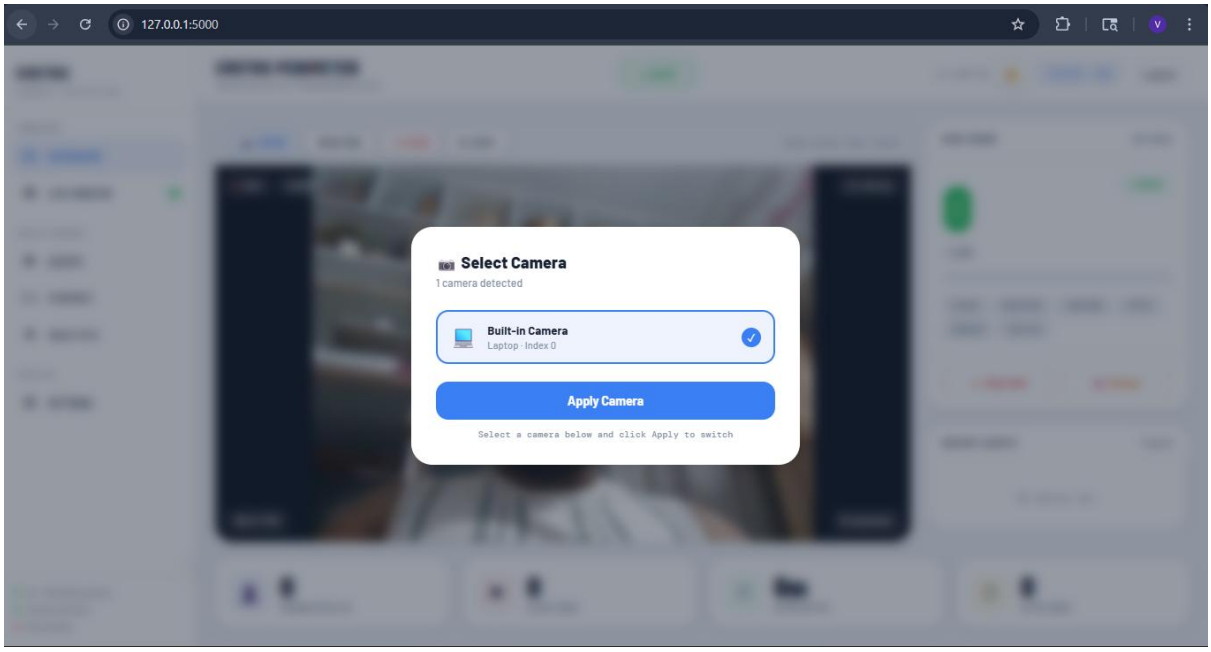
Once logged in, the user is presented with the **CUSTOS Perimeter** dashboard. This command center provides a high-level overview of the security state, including:

- **Live Status**: Real-time health of the system and camera feeds.
- **Quick Stats**: Total persons detected, alerts generated today, and system uptime.
- **Active Zones**: A summary of monitored areas currently under surveillance.



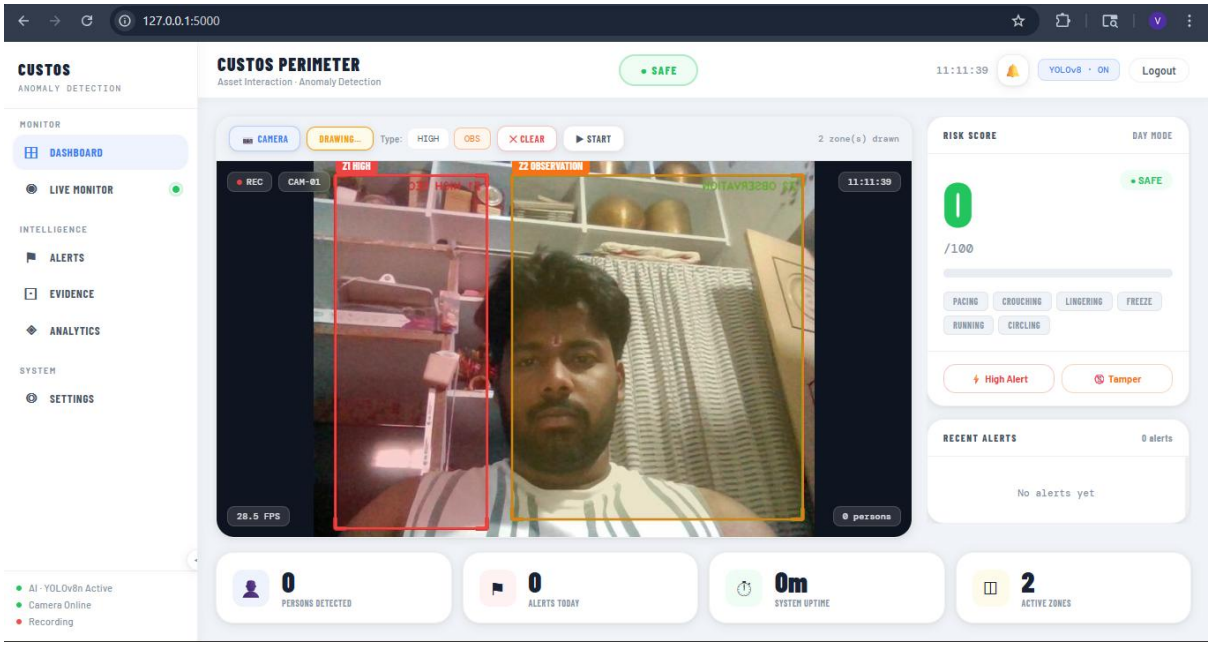
5. Intelligent Camera Integration

CUSTOS is highly flexible, allowing users to select and apply different camera sources. Whether using a built-in laptop camera for local monitoring or external industrial cameras for wider coverage, the integration is seamless.



6. Real-Time Interaction Detection (YOLOv8)

Using the YOLOv8 model, CUSTOS performs real-time object detection and human behavior analysis. The system doesn't just identify a "person" or a "laptop"; it monitors the spatial relationship between them to detect suspicious interactions.

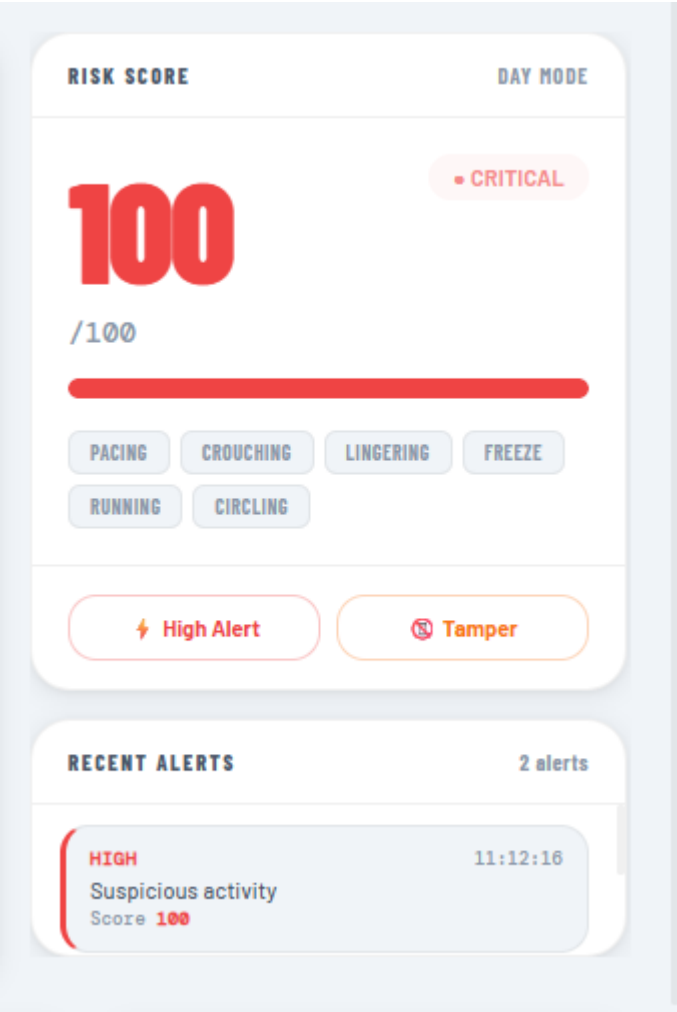


---

7. Advanced Risk Scoring Engine

At the heart of CUSTOS is the **Risk Intelligence Engine**. It evaluates behaviors such as pacing, crouching, lingering, or freezing near a protected asset.

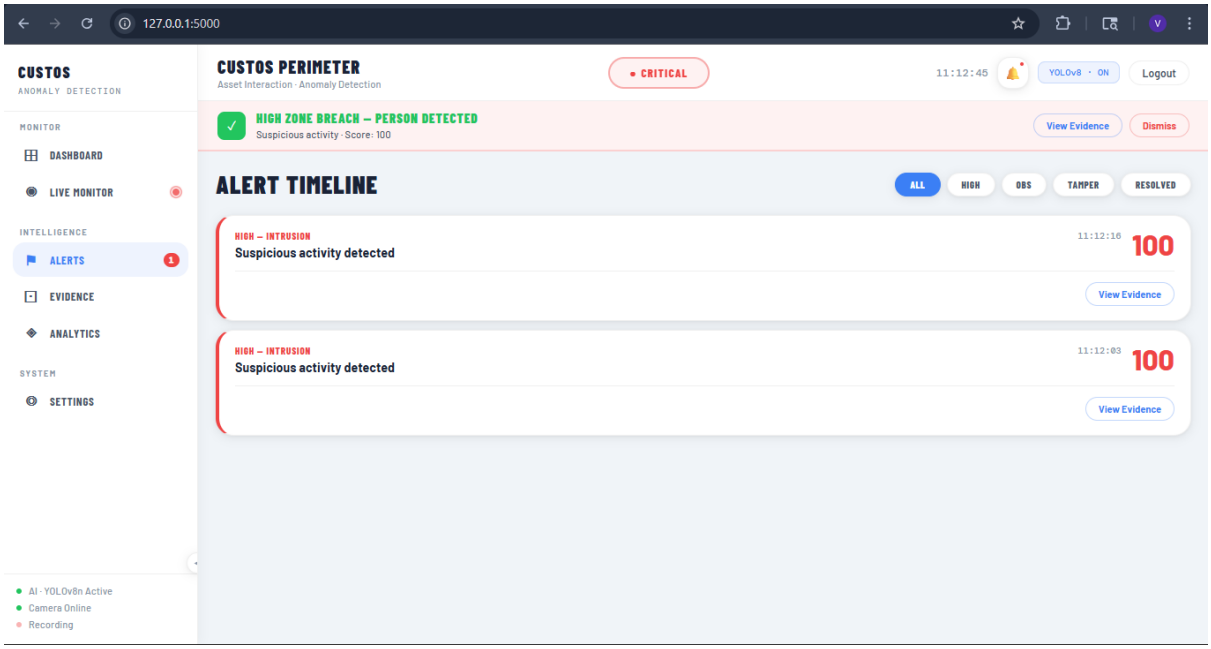
- **Risk Score (0-100)**: A dynamic numerical value representing the current threat level.
- **Behavioral Analysis**: Real-time tracking of specific movement patterns.
- **Alert Status**: Immediate transition to "CRITICAL" when a high-risk interaction occurs.



---

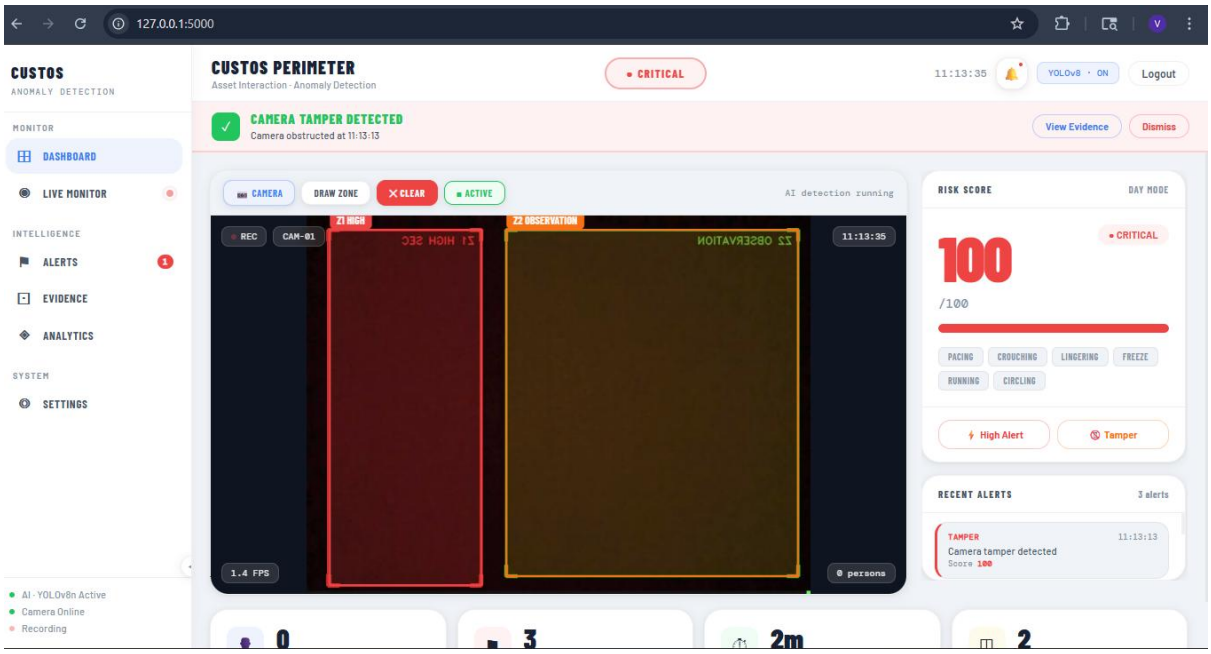
8. Incident Logging and Alert Timeline

Every detected anomaly is logged in the **Alert Timeline**. This provides a chronological record of security breaches, such as "High Zone Breaches" or "Intrusions," allowing security personnel to review past events with full context.



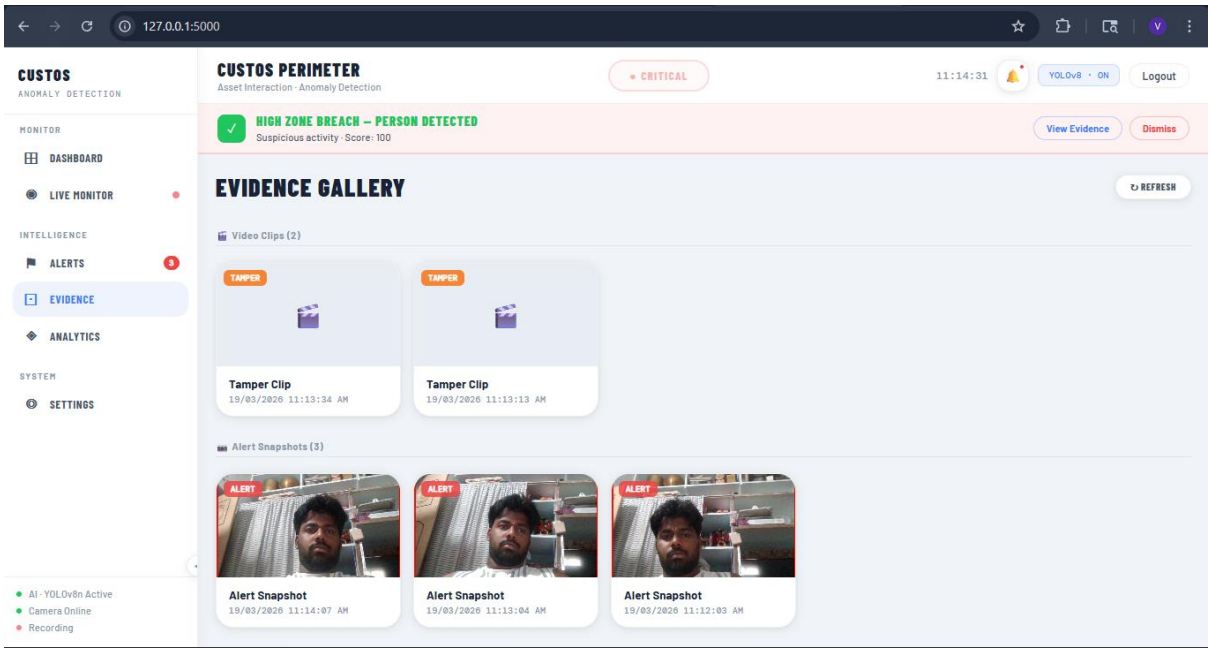
9. Tamper Detection and Evidence Management

CUSTOS includes proactive measures against system interference. If a camera is obstructed or tampered with, the system immediately triggers a "Camera Tamper Detected" alert and captures visual evidence.



10. Evidence Gallery: Snapshots and Video

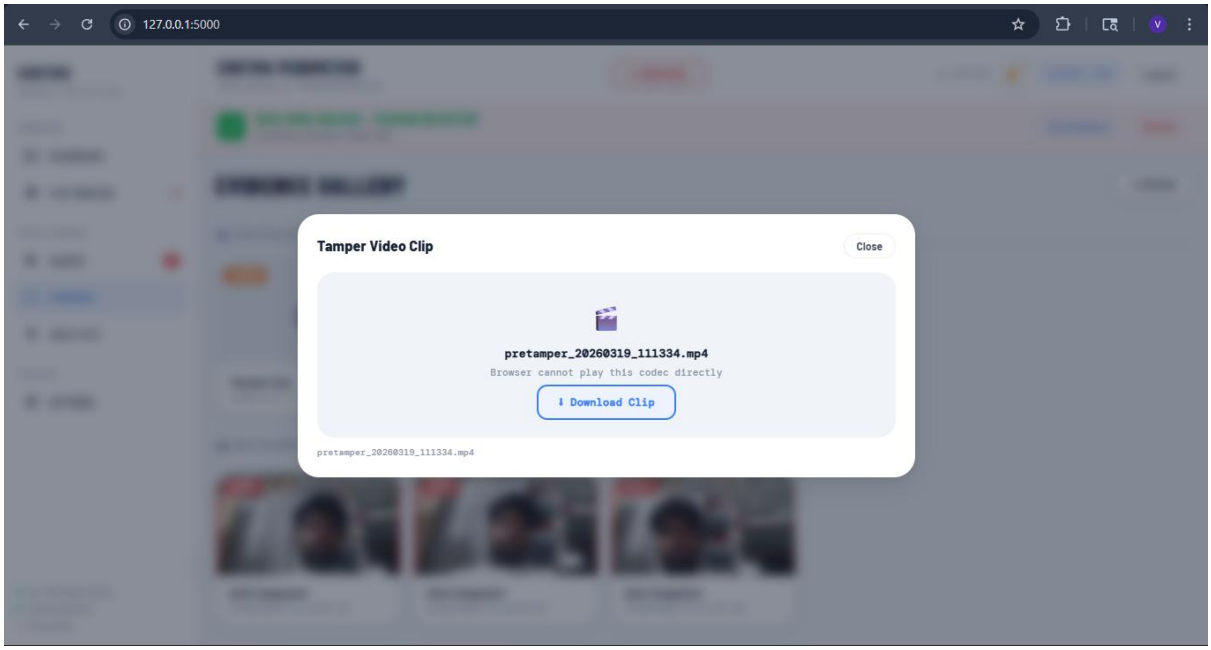
For every critical alert, CUSTOS automatically archives snapshots and video clips. This **Evidence Gallery** serves as a permanent record for legal or internal investigations, ensuring that no suspicious activity goes undocumented.



---

11. Deep Analytics and Operational Insights

Beyond real-time monitoring, CUSTOS provides long-term data analysis. The **Analytics** module tracks risk trends over time, peak alert hours, and average risk scores, helping organizations optimize their security protocols.

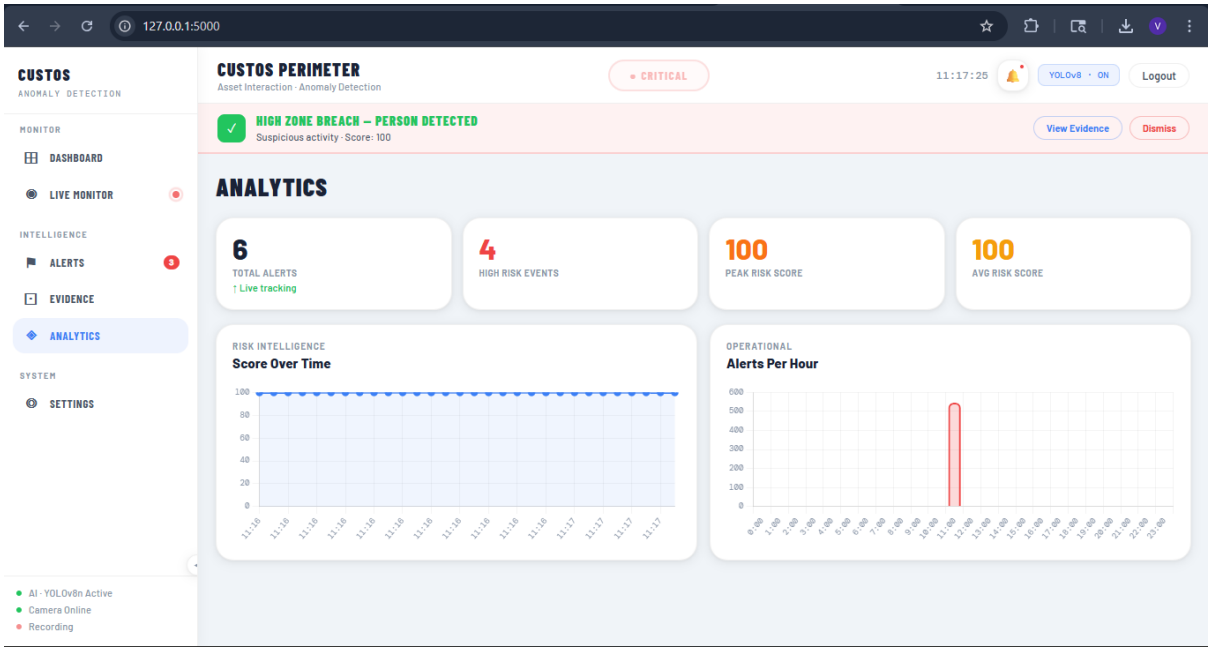


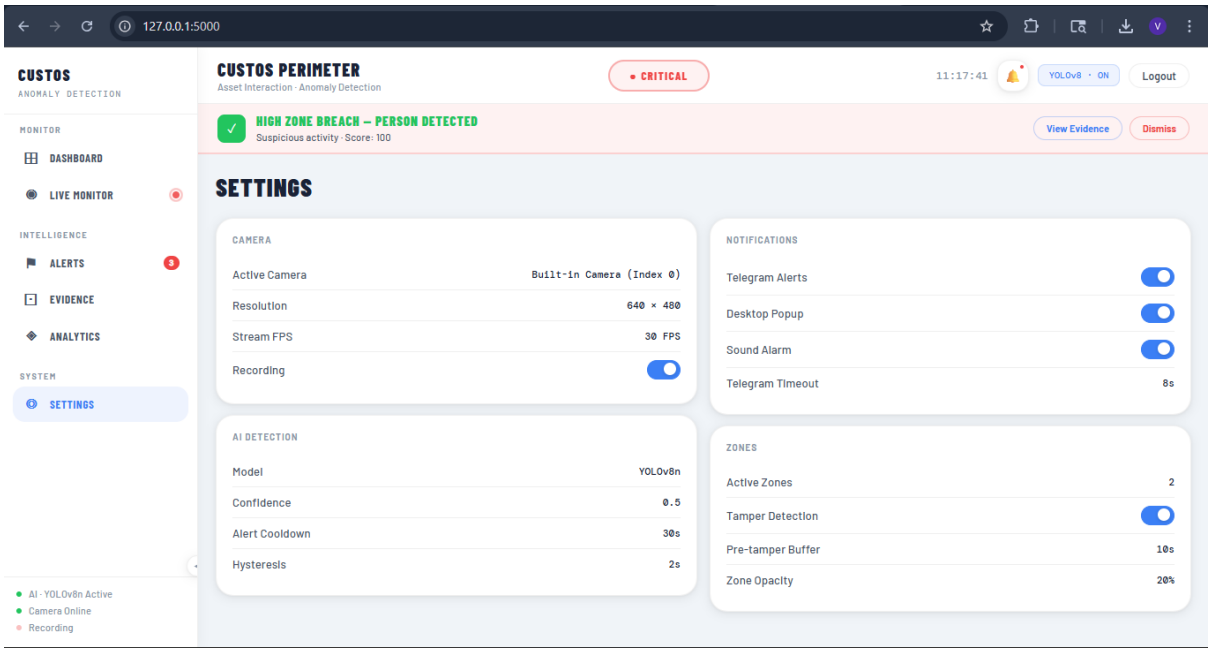
---

12. System Configuration and Customization

The **Settings** panel allows for fine-tuning the AI's sensitivity. Users can adjust:

- **AI Detection**: Model confidence thresholds and alert cooldowns.
- **Notifications**: Telegram alerts, desktop popups, and sound alarms.
- **Zone Management**: Customizing zone opacity and tamper detection sensitivity.





Conclusion

CUSTOS is more than just a camera system; it is a proactive security partner. By combining advanced AI vision with a sophisticated risk engine and a user-friendly management interface, CUSTOS provides the "Security Guard" intelligence needed for modern asset protection.